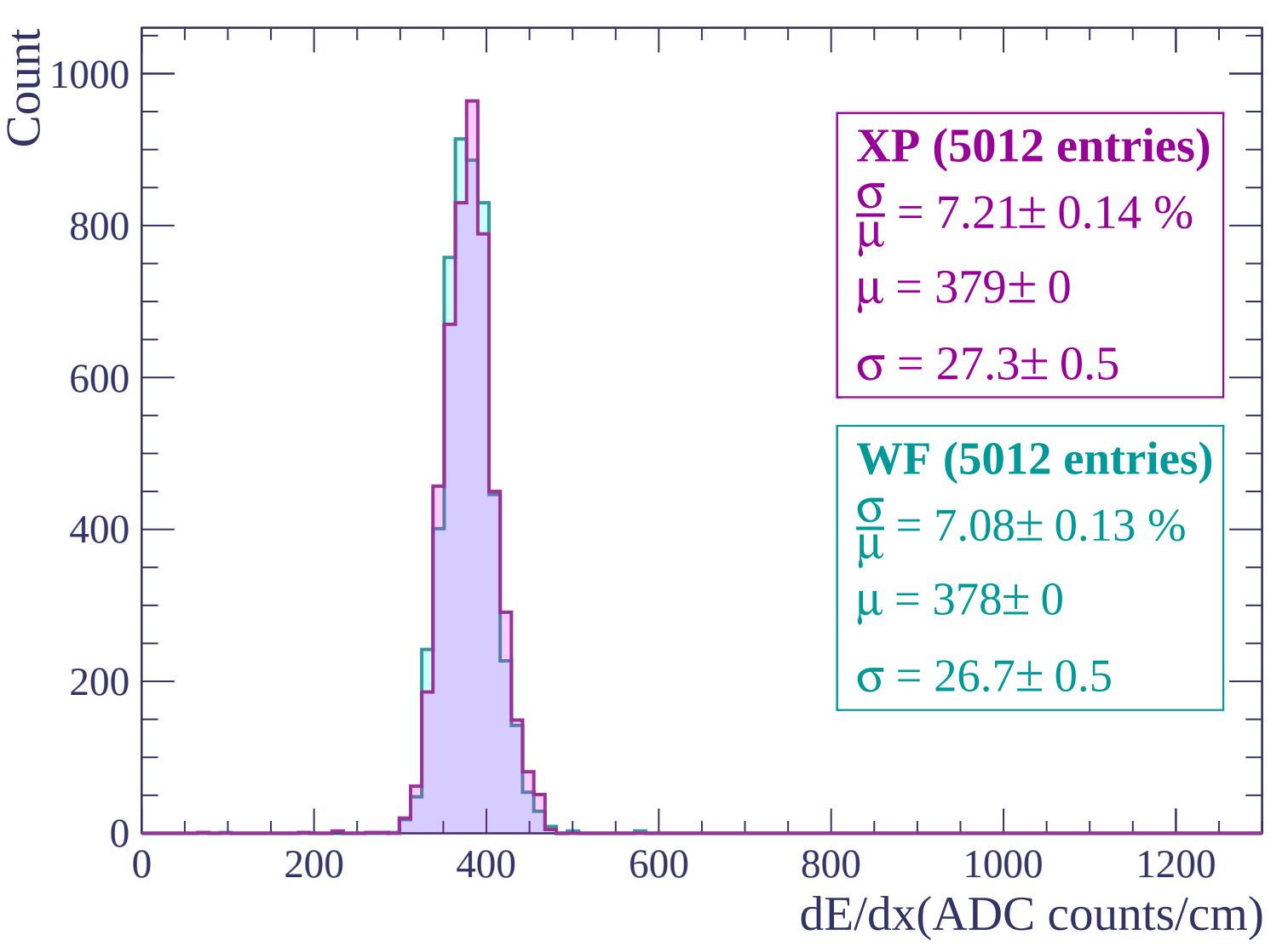
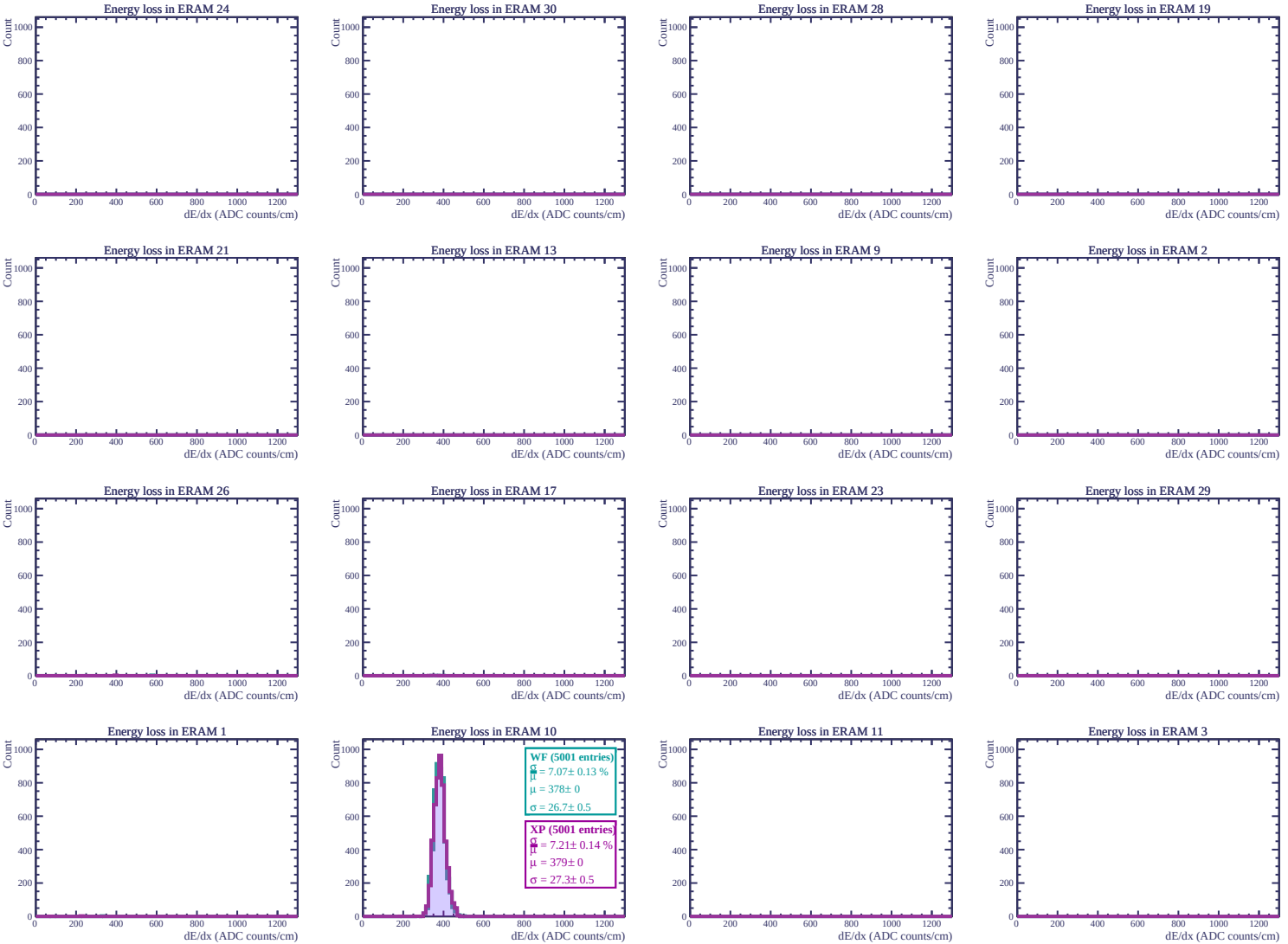
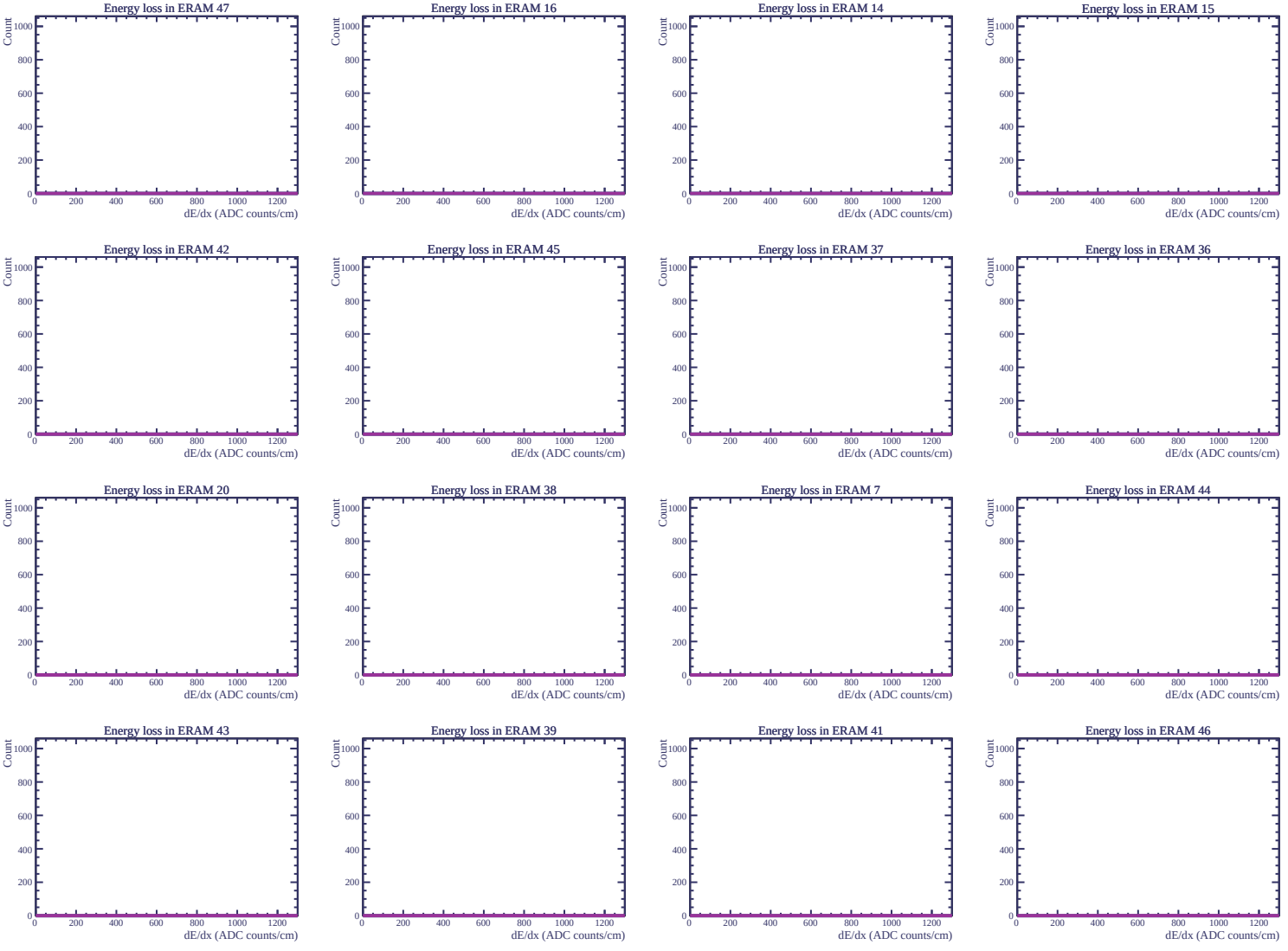
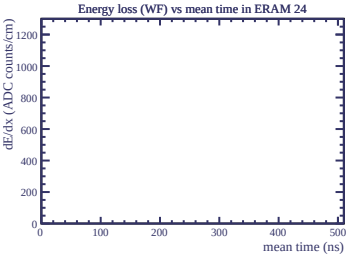
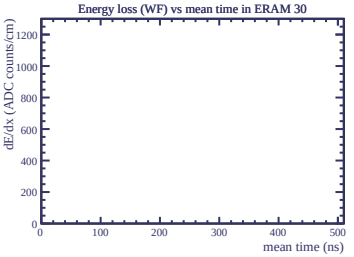
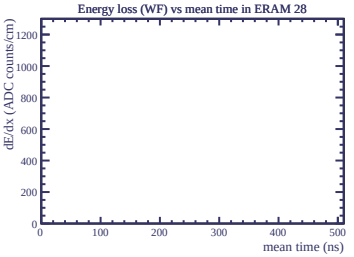

Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (5012 entries)
σ/μ = 7.21± 0.14 %
μ = 379± 0
σ = 27.3± 0.5
WF (5012 entries)
σ/μ = 7.08± 0.13 %
μ = 378± 0
σ = 26.7± 0.5

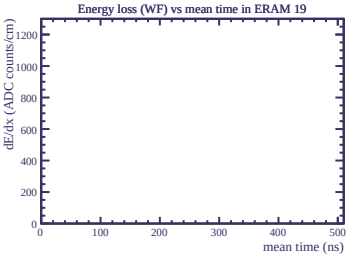

Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
Energy loss in ERAM 13
Energy loss in ERAM 9
Energy loss in ERAM 2
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
Energy loss in ERAM 1
Energy loss in ERAM 10
Energy loss in ERAM 11
Energy loss in ERAM 3
Count
dE/dx (ADC counts/cm)
WF (5001 entries)
σ/μ = 7.07± 0.13 %
μ = 378± 0
σ = 26.7± 0.5
XP (5001 entries)
σ/μ = 7.21± 0.14 %
μ = 379± 0
σ = 27.3± 0.5

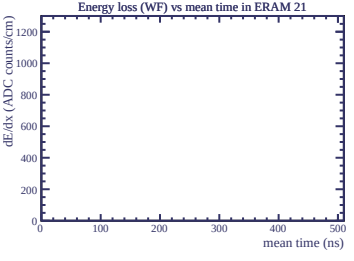

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
Energy loss in ERAM 42
Energy loss in ERAM 45
Energy loss in ERAM 37
Energy loss in ERAM 36
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
Energy loss in ERAM 43
Energy loss in ERAM 39
Energy loss in ERAM 41
Energy loss in ERAM 46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200

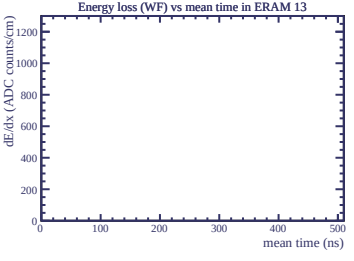

Energy loss (WF) vs mean time in ERAM 24
dE/dx (ADC counts/cm)
mean time (ns)

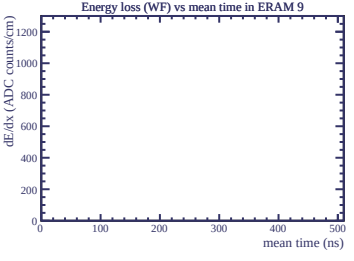

Energy loss (WF) vs mean time in ERAM 30
dE/dx (ADC counts/cm)
mean time (ns)

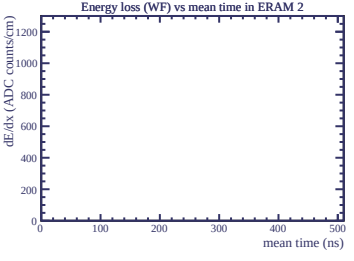

Energy loss (WF) vs mean time in ERAM 28
dE/dx (ADC counts/cm)
mean time (ns)

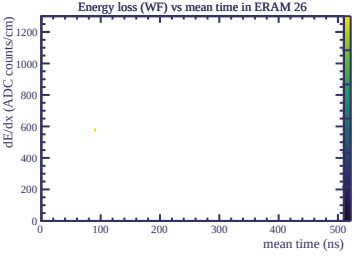

Energy loss (WF) vs mean time in ERAM 19
dE/dx (ADC counts/cm)
mean time (ns)

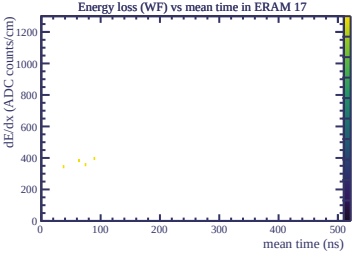

Energy loss (WF) vs mean time in ERAM 21
dE/dx (ADC counts/cm)
mean time (ns)

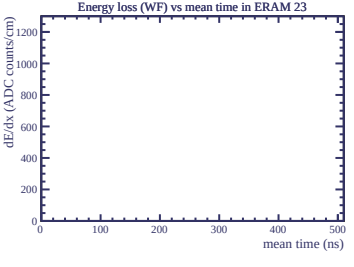

Energy loss (WF) vs mean time in ERAM 13
dE/dx (ADC counts/cm)
mean time (ns)

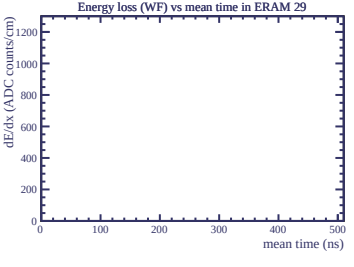

Energy loss (WF) vs mean time in ERAM 9
dE/dx (ADC counts/cm)
mean time (ns)

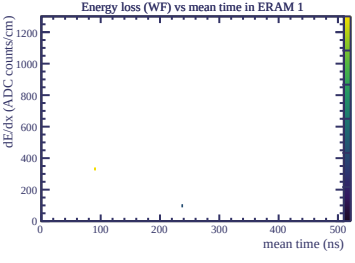

Energy loss (WF) vs mean time in ERAM 2
dE/dx (ADC counts/cm)
mean time (ns)

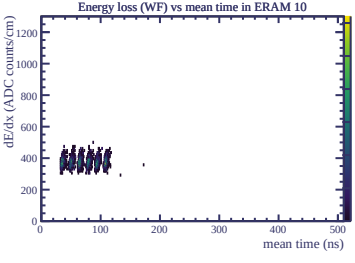

Energy loss (WF) vs mean time in ERAM 26
dE/dx (ADC counts/cm)
mean time (ns)

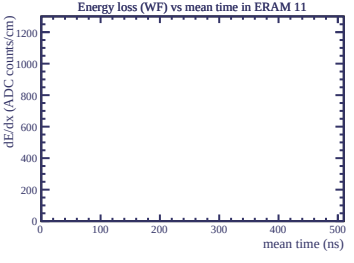

Energy loss (WF) vs mean time in ERAM 17
dE/dx (ADC counts/cm)
mean time (ns)

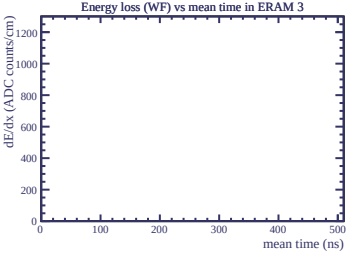

Energy loss (WF) vs mean time in ERAM 23
dE/dx (ADC counts/cm)
mean time (ns)

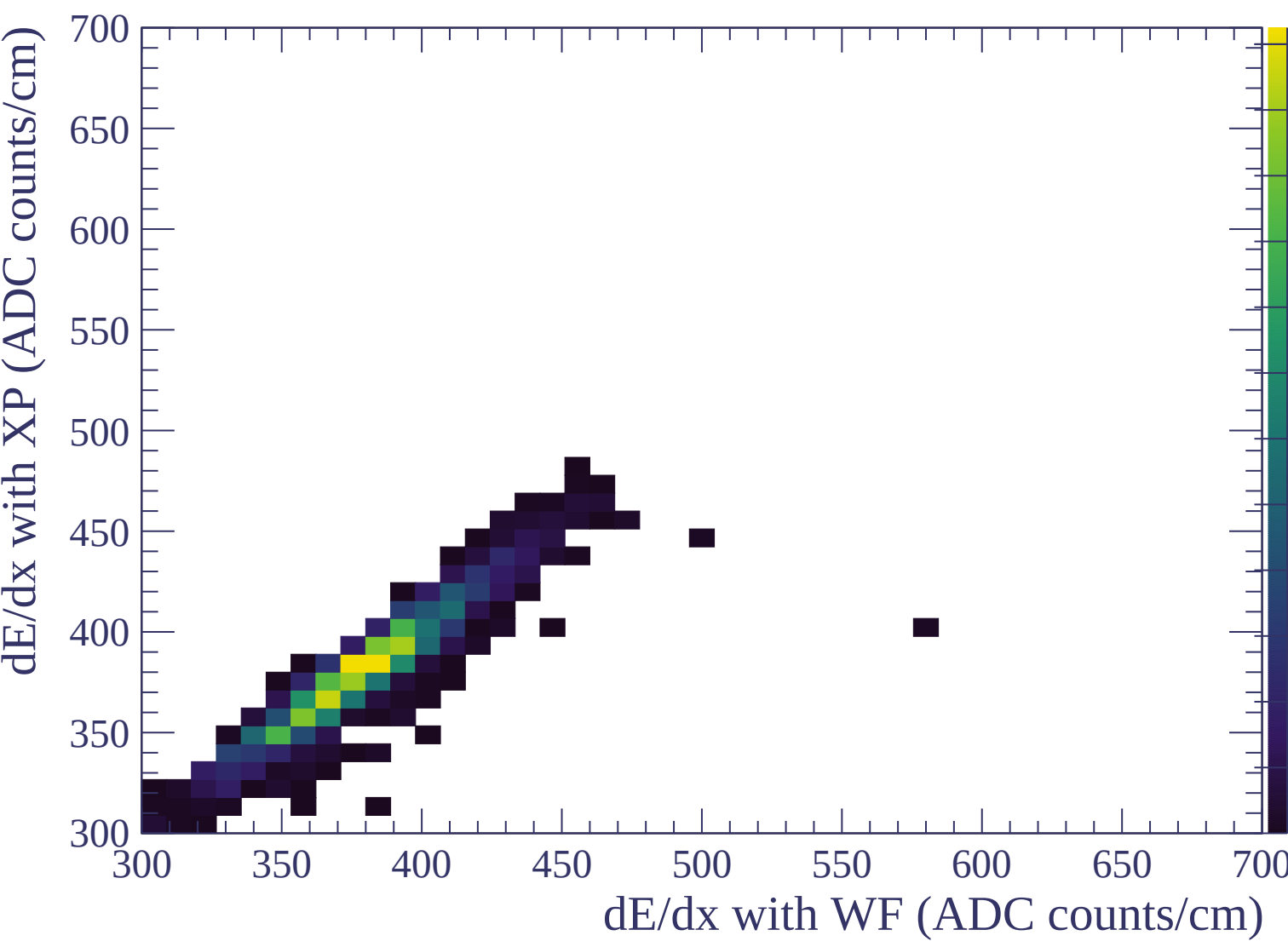

Energy loss (WF) vs mean time in ERAM 29
dE/dx (ADC counts/cm)
mean time (ns)

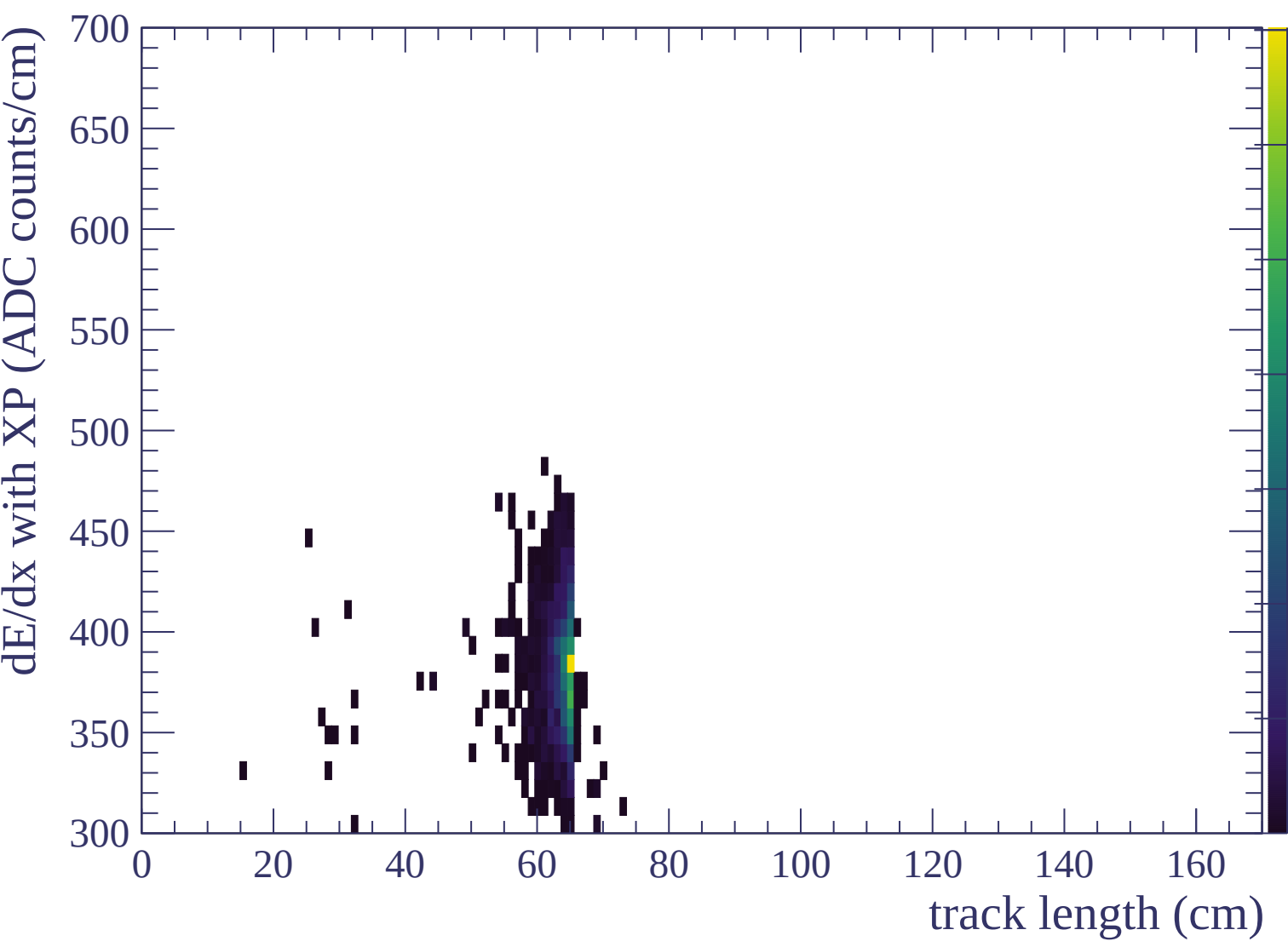

Energy loss (WF) vs mean time in ERAM 1
dE/dx (ADC counts/cm)
mean time (ns)

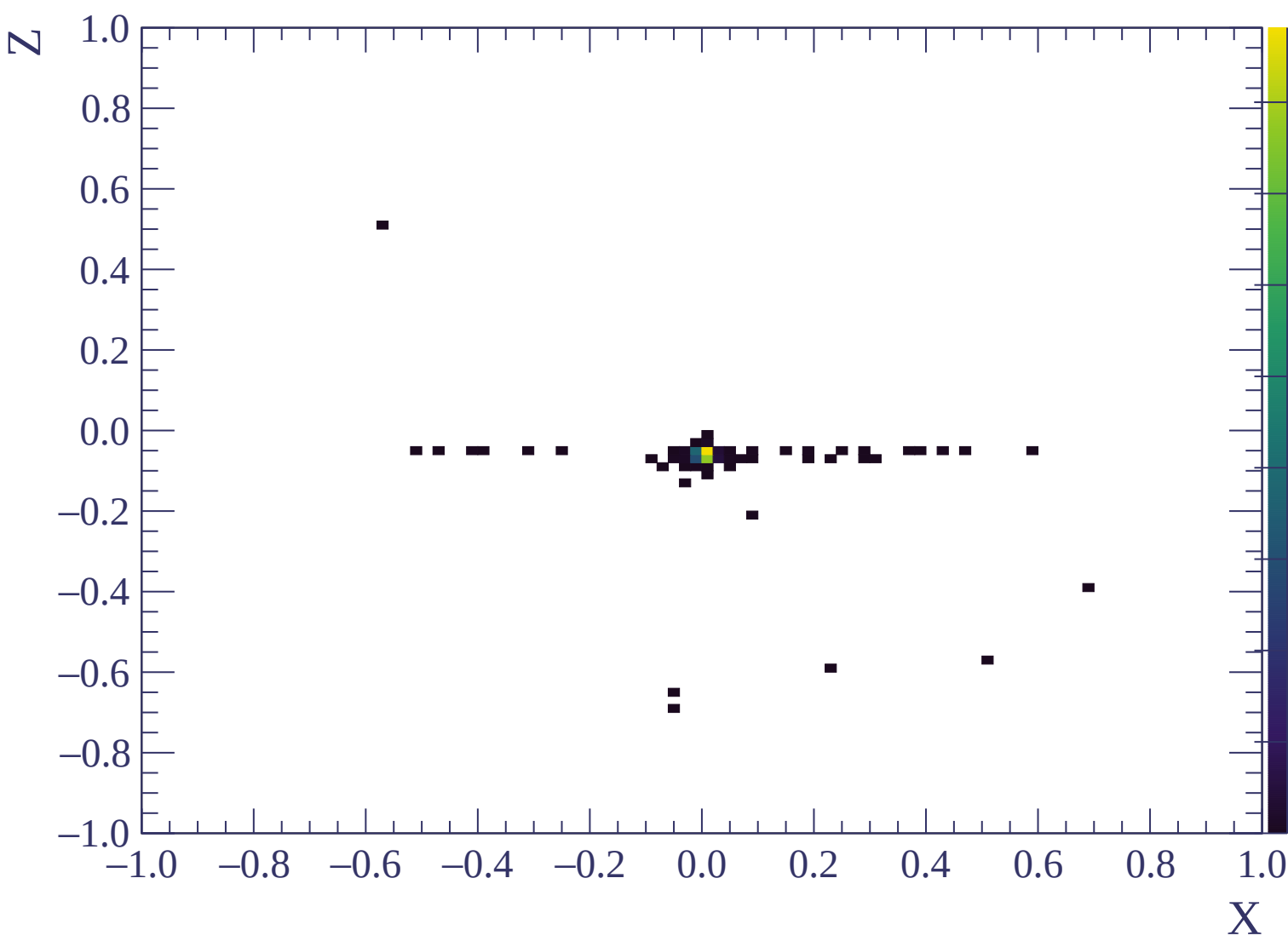

Energy loss (WF) vs mean time in ERAM 10
dE/dx (ADC counts/cm)
mean time (ns)

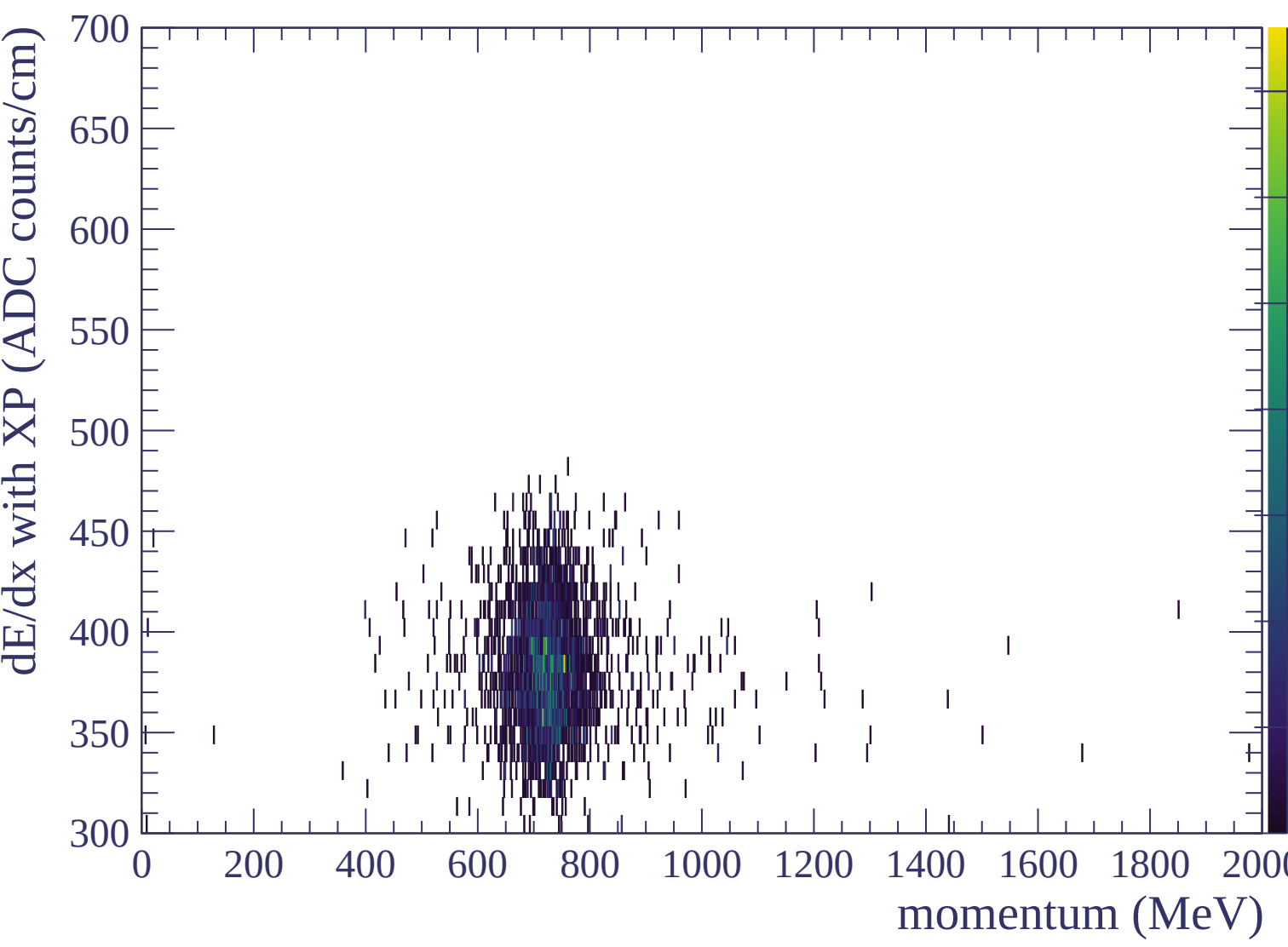

Energy loss (WF) vs mean time in ERAM 11
dE/dx (ADC counts/cm)
mean time (ns)

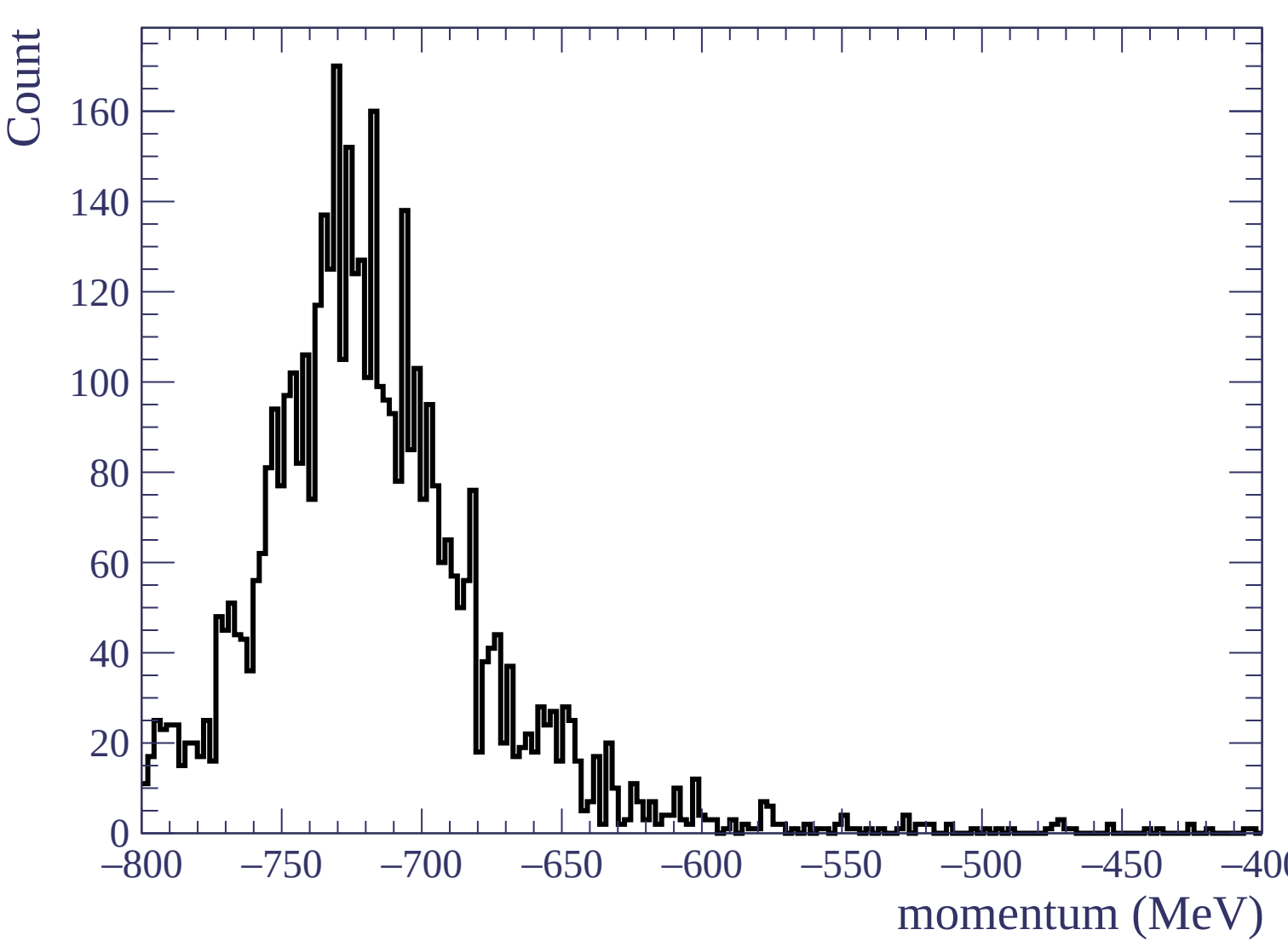

Energy loss (WF) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

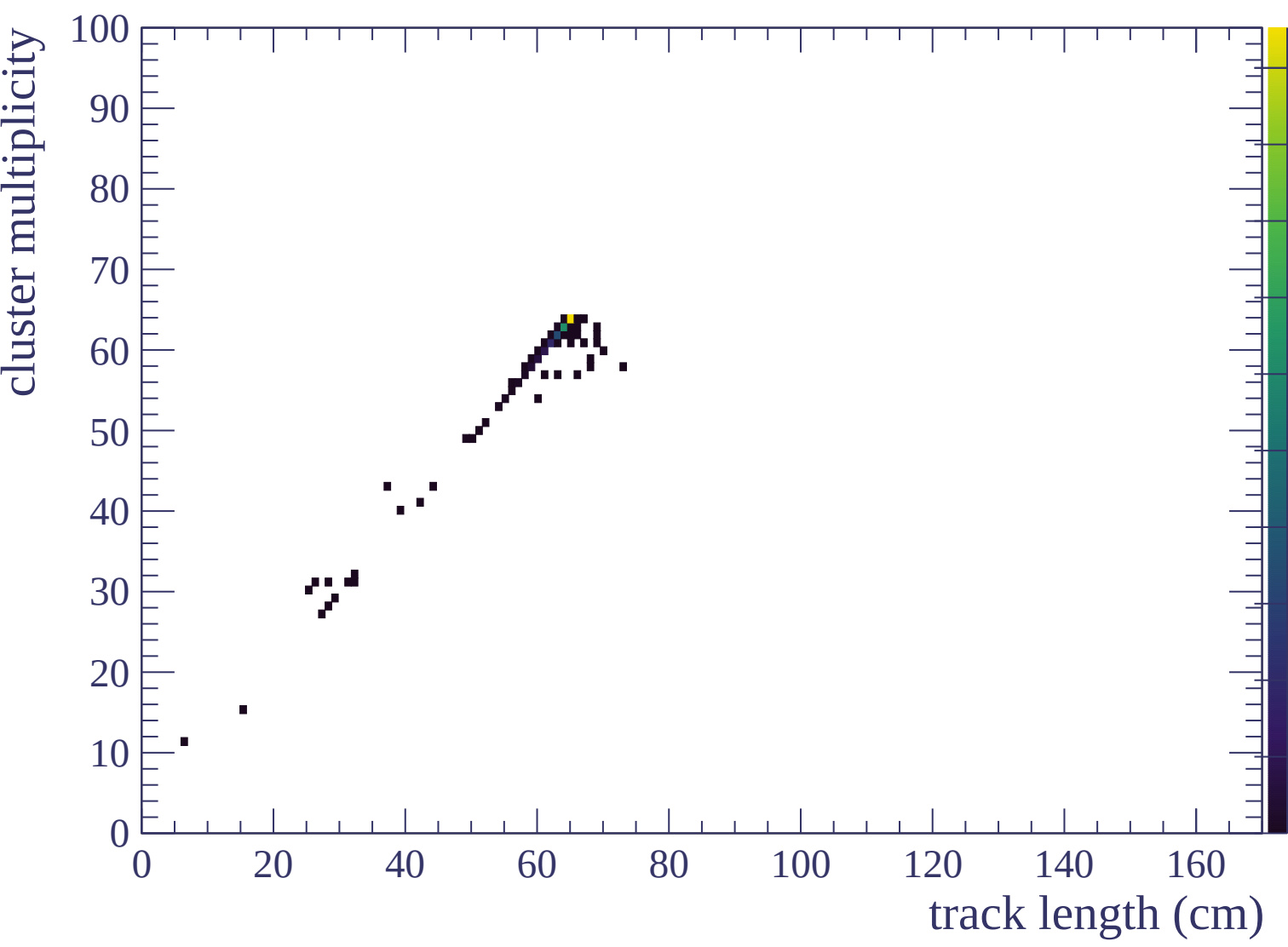

dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
300
350
400
450
500
550
600
650
700

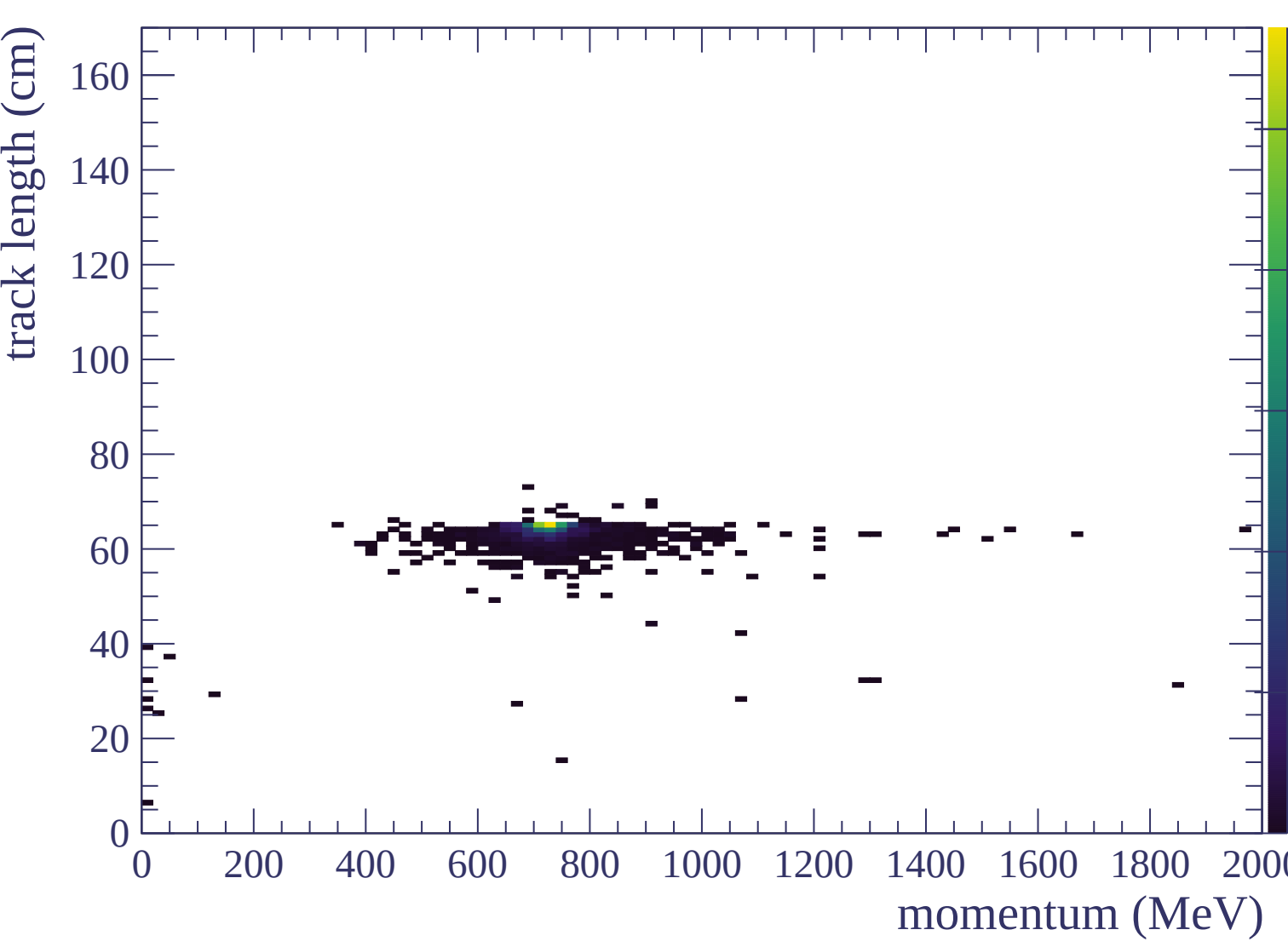

dE/dx with XP (ADC counts/cm)
700
650
600
550
500
450
400
350
300
0
20
40
60
80
100
120
140
160
track length (cm)

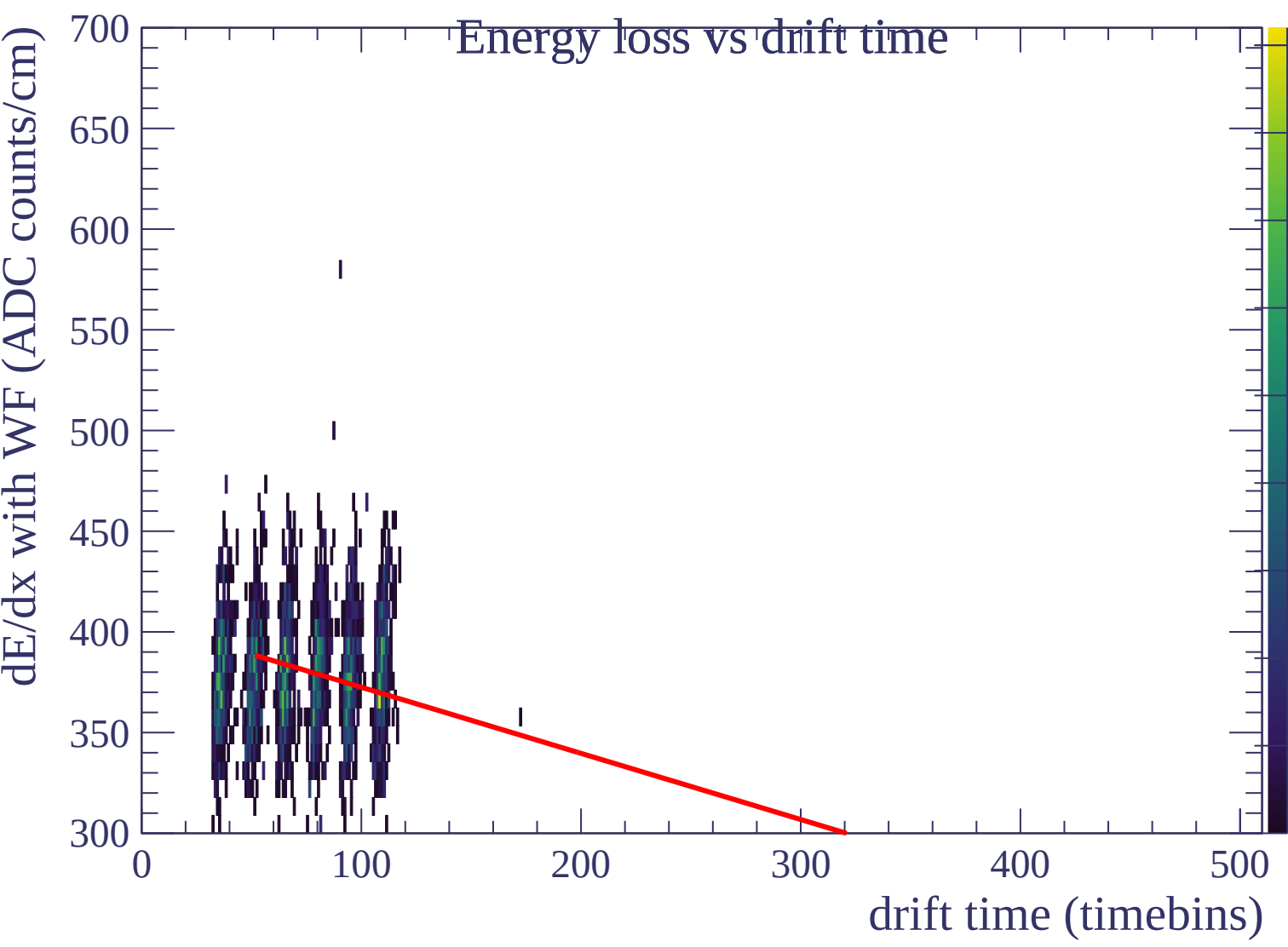

Z
1.0
0.8
0.6
0.4
0.2
0.0
–0.2
–0.4
–0.6
–0.8
–1.0
–1.0
–0.8
–0.6
–0.4
–0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

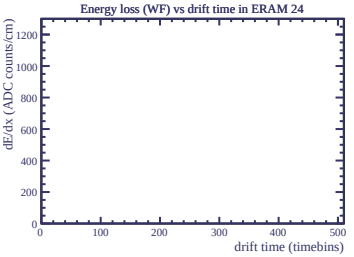

dE/dx with XP (ADC counts/cm)
700
650
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000
momentum (MeV)

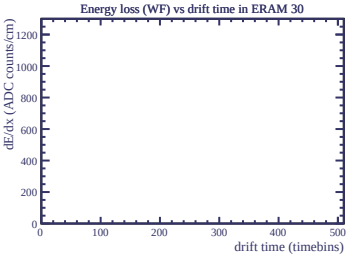

Count
160
140
120
100
80
60
40
20
0
–800
–750
–700
–650
–600
–550
–500
–450
momentum (MeV)

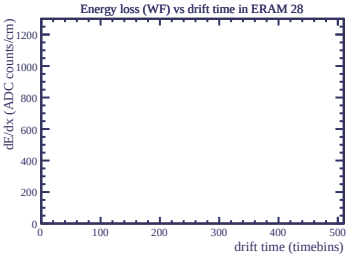

cluster multiplicity
track length (cm)
0
10
20
30
40
50
60
70
80
90
100
0
20
40
60
80
100
120
140
160

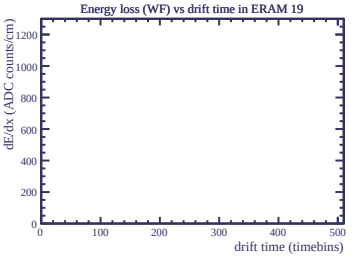

track length (cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
1400
1600
1800
2000
momentum (MeV)

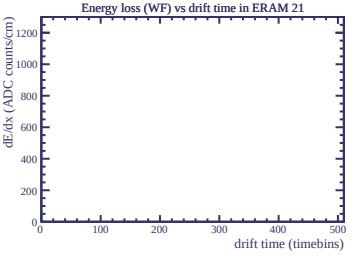

Energy loss vs drift time
dE/dx with WF (ADC counts/cm)
700
650
600
550
500
450
400
350
300
0
100
200
300
400
500
drift time (timebins)

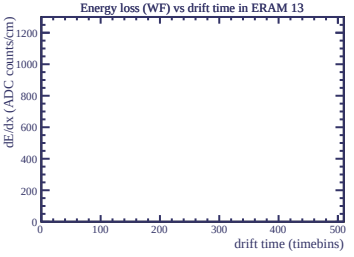

Energy loss (WF) vs drift time in ERAM 24
dE/dx (ADC counts/cm)
drift time (timebins)

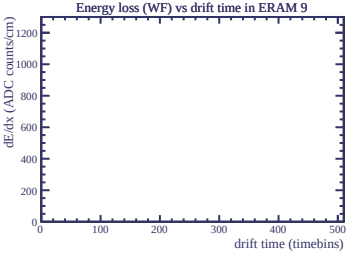

Energy loss (WF) vs drift time in ERAM 30
dE/dx (ADC counts/cm)
drift time (timebins)

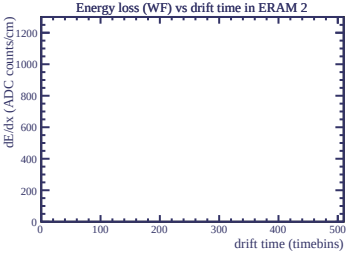

Energy loss (WF) vs drift time in ERAM 28
dE/dx (ADC counts/cm)
drift time (timebins)

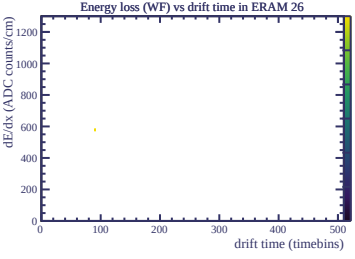

Energy loss (WF) vs drift time in ERAM 19
dE/dx (ADC counts/cm)
drift time (timebins)

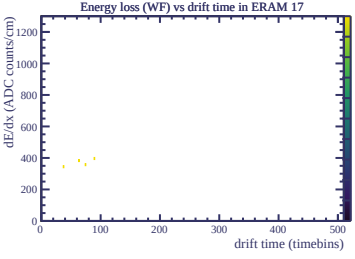

Energy loss (WF) vs drift time in ERAM 21
dE/dx (ADC counts/cm)
drift time (timebins)

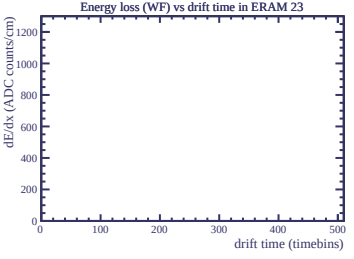

Energy loss (WF) vs drift time in ERAM 13
dE/dx (ADC counts/cm)
drift time (timebins)

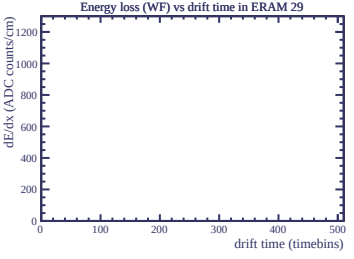

Energy loss (WF) vs drift time in ERAM 9
dE/dx (ADC counts/cm)
drift time (timebins)

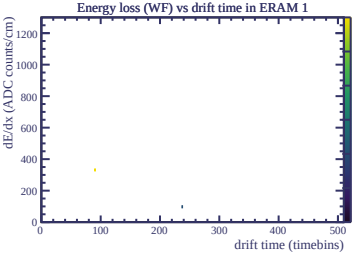

Energy loss (WF) vs drift time in ERAM 2
dE/dx (ADC counts/cm)
drift time (timebins)

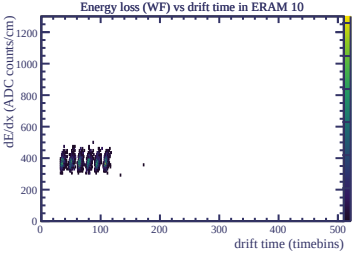

Energy loss (WF) vs drift time in ERAM 26
dE/dx (ADC counts/cm)
drift time (timebins)

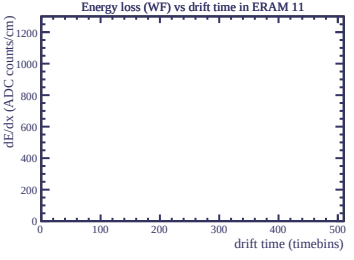

Energy loss (WF) vs drift time in ERAM 17
dE/dx (ADC counts/cm)
drift time (timebins)

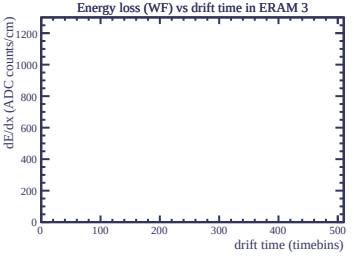

Energy loss (WF) vs drift time in ERAM 23
dE/dx (ADC counts/cm)
drift time (timebins)

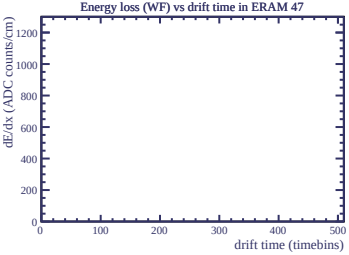
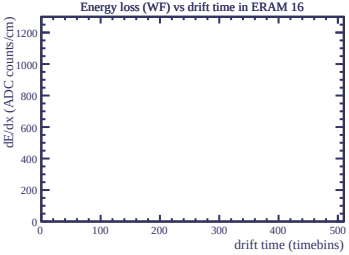
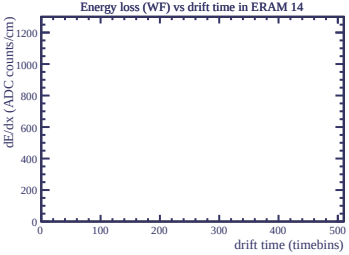
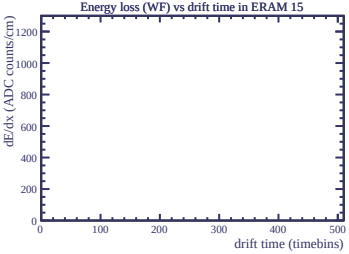
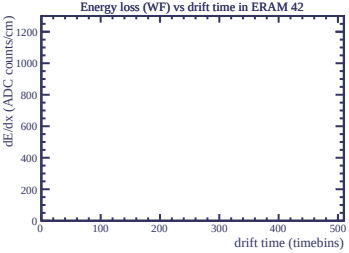
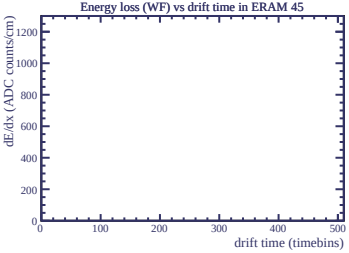
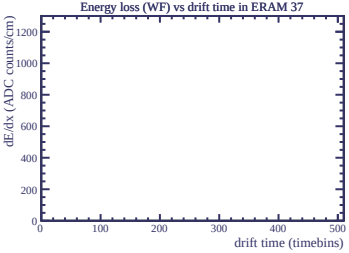
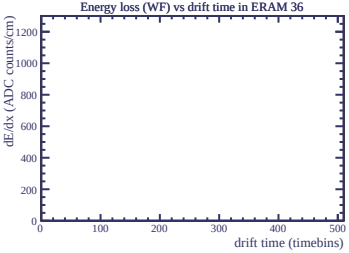
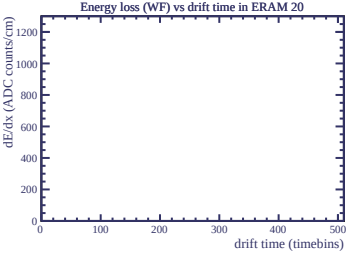
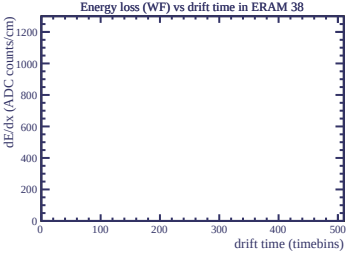
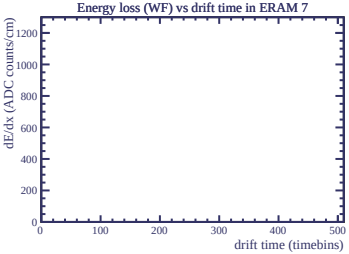
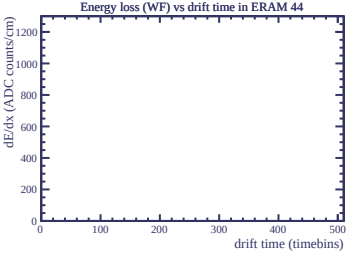
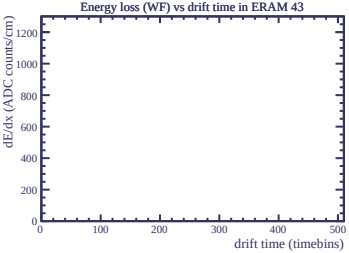
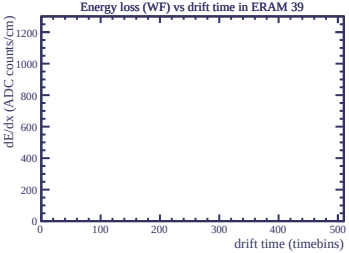
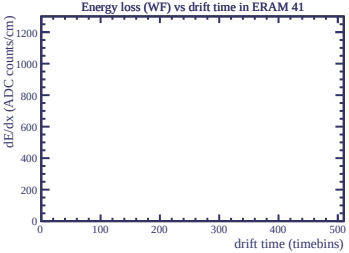
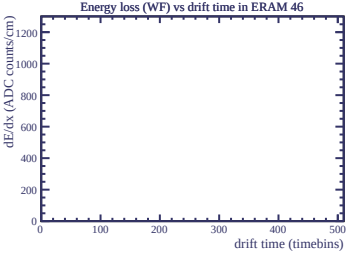

Energy loss (WF) vs drift time in ERAM 29
dE/dx (ADC counts/cm)
drift time (timebins)

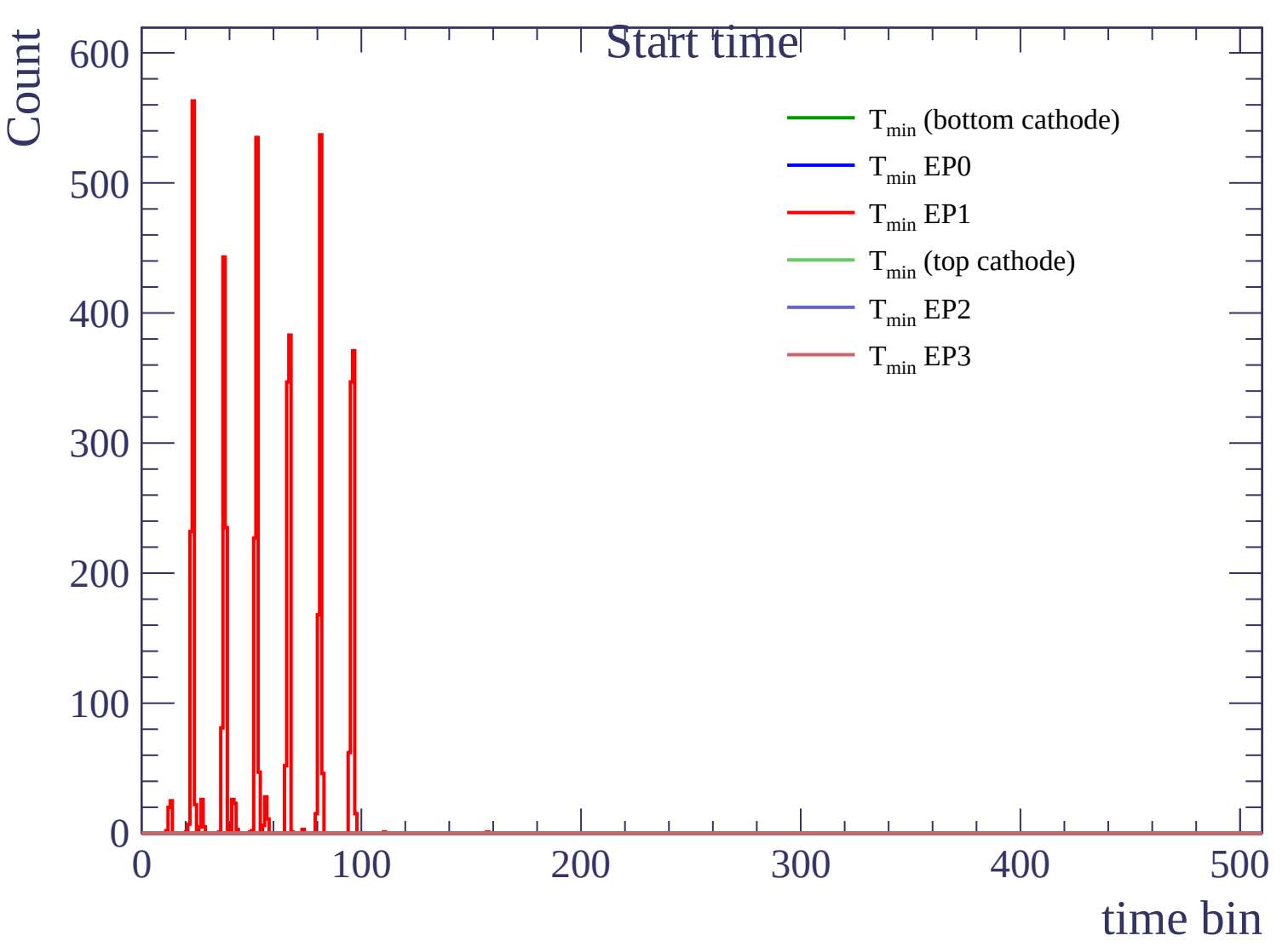

Energy loss (WF) vs drift time in ERAM 1
dE/dx (ADC counts/cm)
drift time (timebins)

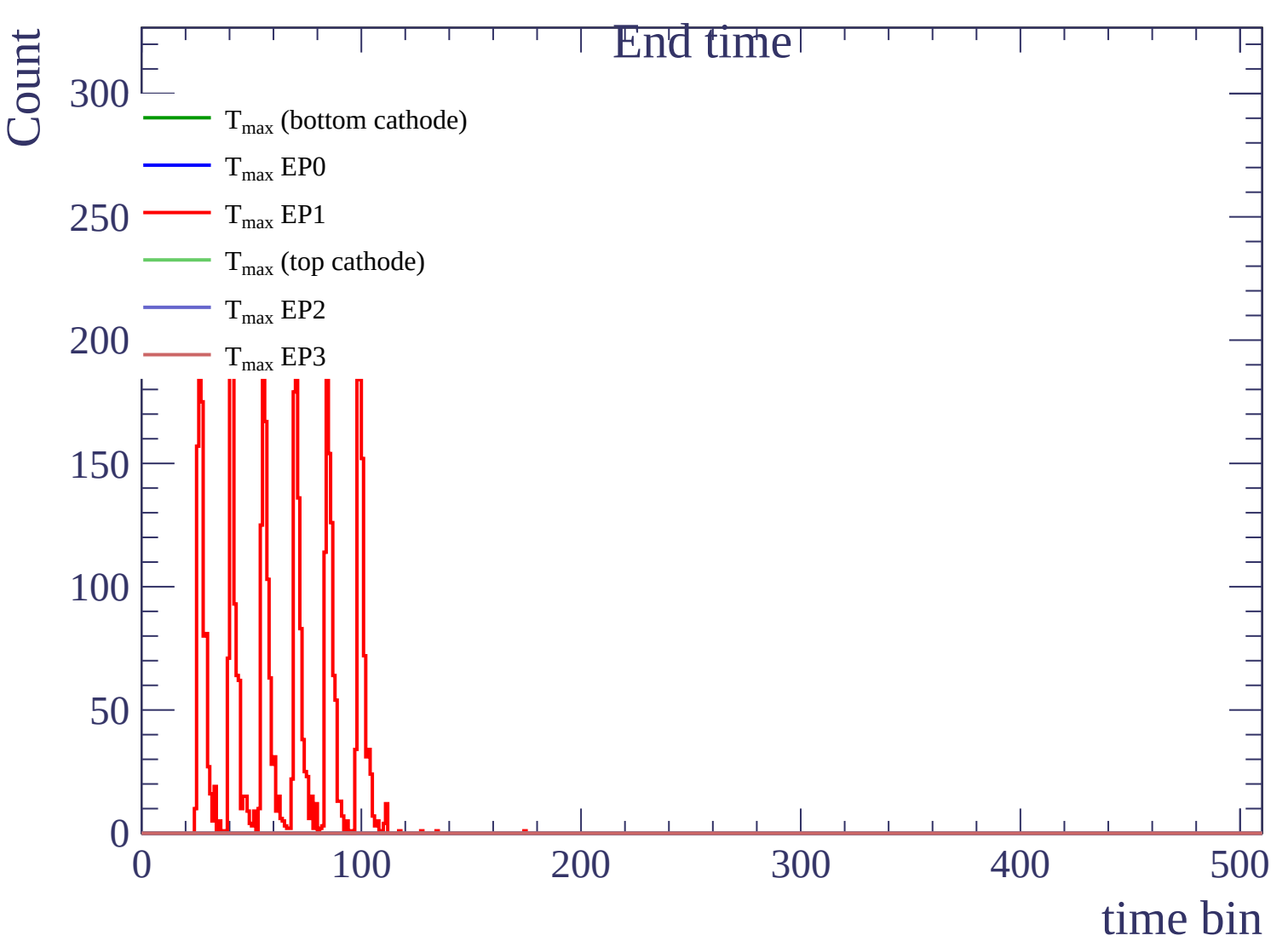

Energy loss (WF) vs drift time in ERAM 10
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (WF) vs drift time in ERAM 11
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (WF) vs drift time in ERAM 3
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (WF) vs drift time in ERAM 47
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 16
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 14
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 15
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 42
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 45
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 37
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 36
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 20
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 38
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 7
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 44
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 43
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 39
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 41
dE/dx (ADC counts/cm)
drift time (timebins)

Energy loss (WF) vs drift time in ERAM 46
dE/dx (ADC counts/cm)
drift time (timebins)


Start time
Count
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3
0
100
200
300
400
500
600


End time
Count
300
250
200
150
100
50
0
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
100
200
300
400
500
time bin